

BRAIN TUMOR DETECTION IN PEDIATRIC MRI

COMPUTER VISION PROJECT REPORT

BraTS-PEDs Dataset

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1. INTRODUCTION

Brain tumors are one of the most serious and life threatening conditions that humans are affected by. Detection of brain tumor at early stages through MRI scans can help the patients and medical professionals can treat them timely. However, manual review of several MRI scans is inefficient, time-taking and requires thorough medical expert opinions and knowledge.

Computer vision helps us build ways to automate this manual process by allowing machines to analyze brain scans, clean the scans, identify tumor based regions and classifying them into different classes so they can be treated accordingly.

This computer vision project is based on working on BraTS-PEDs (Brain Tumor Segmentation - Pediatric) Dataset, that contains 25 patients and their brain MRI scans, with 4 different modalities and one segmentation mask. So each patient file contains 5 images (4 brain modalities, 1 segmentation mask). the modalities we have worked with are: T1n (native), T1c (contrast), T2w (weighted), and T2-Flair. These four modalities capture the brain regions differently ultimately giving the complete picture.

The project pipeline starts from batch processing of the scans of all the patients to using a deep learning approach to identify the tumor region in the brain scans.

2. LITERATURE REVIEW

Brain tumor segmentation from MRI scans is a broad and extensive research area. Traditionally, the methods used to identify tumor regions were by manual inspection or using rule based systems, or applying filters for noise removal and then applying a baseline machine learning model to classify the whole MRI scan as tumor or no-tumor scan.

In this project's pipeline, we covered image filtering like mean, median, and Gaussian filter and applying morphological operations to work only on the region of interest. Image filtering is guaranteed to remove noise from scans. Through chain coding and convex hull representation of the tumors in the brain, these methods are used for boundary representation and shape description of the brain in computer vision.

A research study mainly focusing on six classes of tumors [1], used five computer vision classifiers, MLP, BN, PART, RFC and these classifiers were deployed in an optimized statistical multi-free dataset. The MLP classifier with fine-tuned hyper-parameters demonstrated exceptionally high accuracy compared to the others.

A deep learning model, VGG16, was utilized to implement an effective, fast, and easy way to identify whether MRI scans have tumors or not. The model was created from scratch, and it reached a train accuracy of 100% and validation accuracy of 99.28% which shows that the model successfully detected the tumor regions and classified it into any of the one classes [2].

3. PROBLEM STATEMENT

The goal of this project is build a complete pipeline that takes raw Pediatric brain MRI scans as input and then applying multiple operations like filtering, bit-depth detection, noise removal (morphological operations), edge detection filters etc.

MRI scans contain salt and pepper noise, that should be removed through morphological operations. The brain tumor regions do not have clear boundaries defined by pixels, for that we use edge detection filters. We computed different features based on texture and shape of the tumor, so we can classify them as malignant or benign.

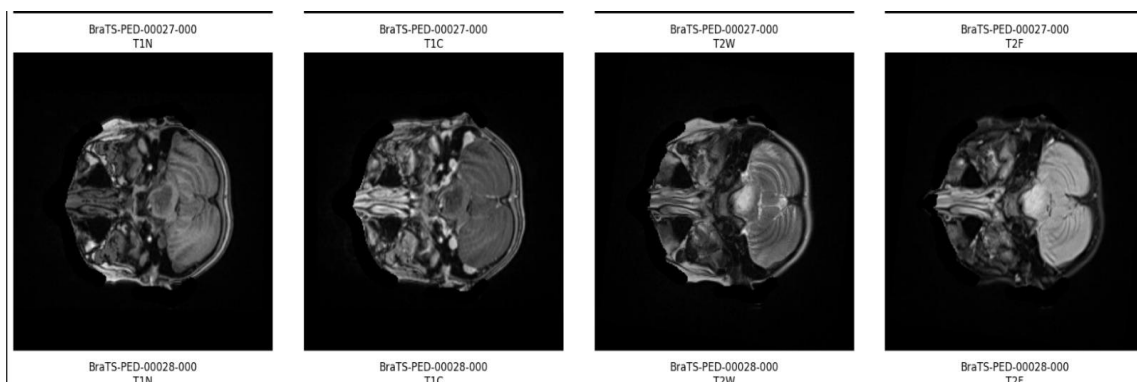
4. METHODOLOGY

The full project pipeline is divided among 4 assignments. Each one uses the output of the previous as its input.

4.1 Assignment 01 - Image Preprocessing

4.1.1 Batch processing:

Each patient folder contains a 3D MRI volume with 155 axial slices. The tumor regions are not available in all the slices, for every patient the tumor region slices are different. We used the T2-FLAIR slice because it is the volume that highlights tumor regions clearly making it the the most crucial modality for this project. We calculated tissue intensity threshold and then for slice searching we excluded the lower face/neck and top-scalp artifacts. The algorithms selects the pixels with the highest intensity variance among pixels in order to meet the minimum brain coverage threshold.



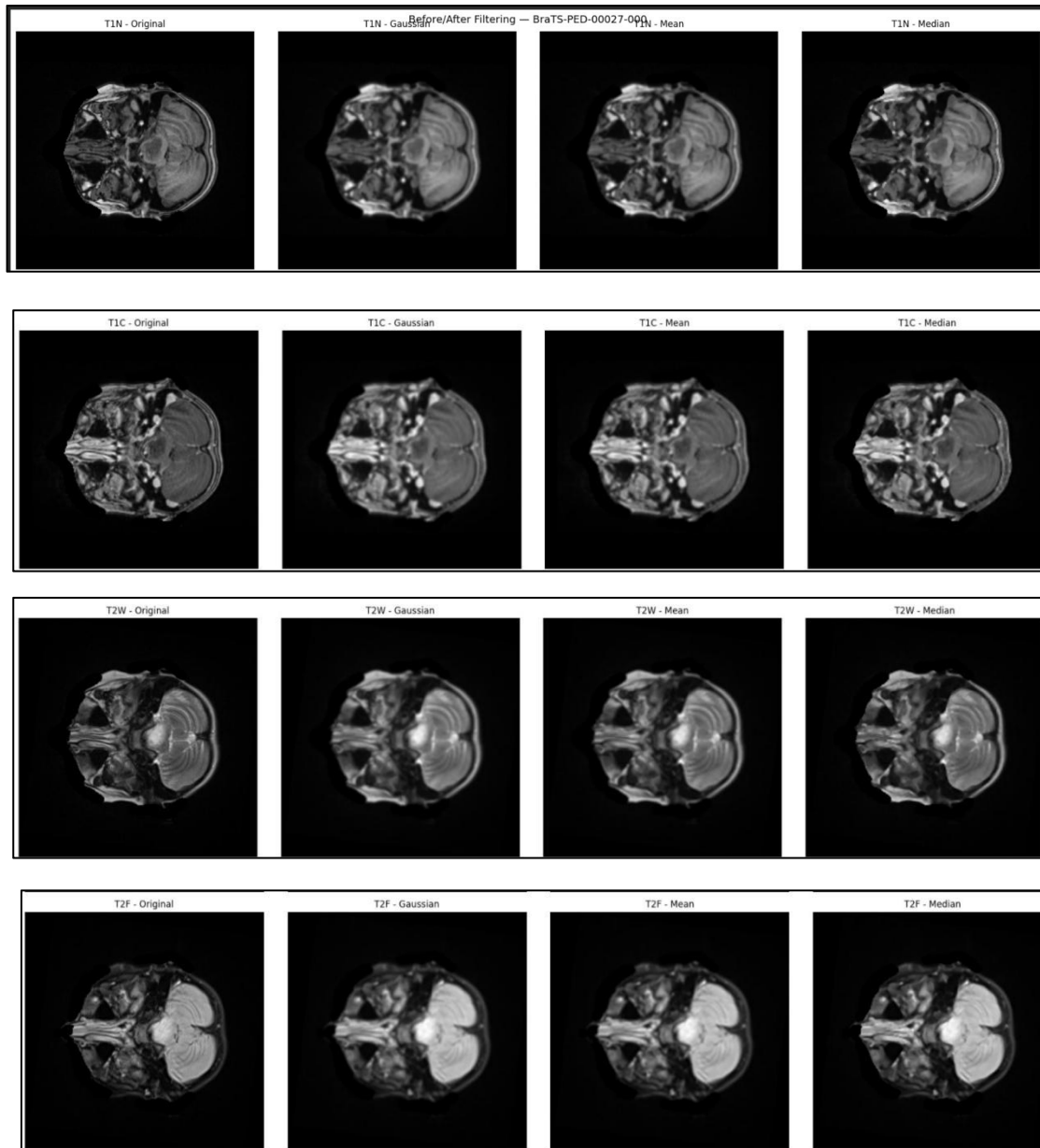
4.1.2 Radiometric Resolution (Bit-Depth Detection):

After slice selection, we calculated bit-depth across all four modalities. If pixel values are in the range of 255. it is an 8-bit image, if the range is upto 65535 then it is a 16-bit image and so on. Accurately calculating the bit-depth saves

from false contouring, a concept where an 8-bit image is mistaken as a 16-bit image, resulting in fake edges on smooth regions of the scan. All the scans are 16-bit images.

4.1.3 Filtering:

Three standard spatial filters were applied to each modality for every patient. The kernel size was kept 3x3. Each filter is applied to the normalized the image. Median filter produces the best results among all three.



4.1.4 PSNR and SSIM:

PSNR shows how much the image was distorted after filter application. The higher PSNR score means image was not distorted a lot. With values above 30 db, the image is considered good. SSIM measures whether the structure of the image is retained, values close to 1 mean that structure was intact.

These evaluations were done for all three filters across all 25 patients and their 4 modalities. Median filter gave the best evaluations cores for both metrics.

Patient	Modality	Filter	PSNR (dB)	SSIM
=====				
BraTS-PED-00026-000	t1n	gaussian	28.8457	0.9382
BraTS-PED-00026-000	t1n	mean	29.3442	0.9452
BraTS-PED-00026-000	t1n	median	31.8605	0.9646

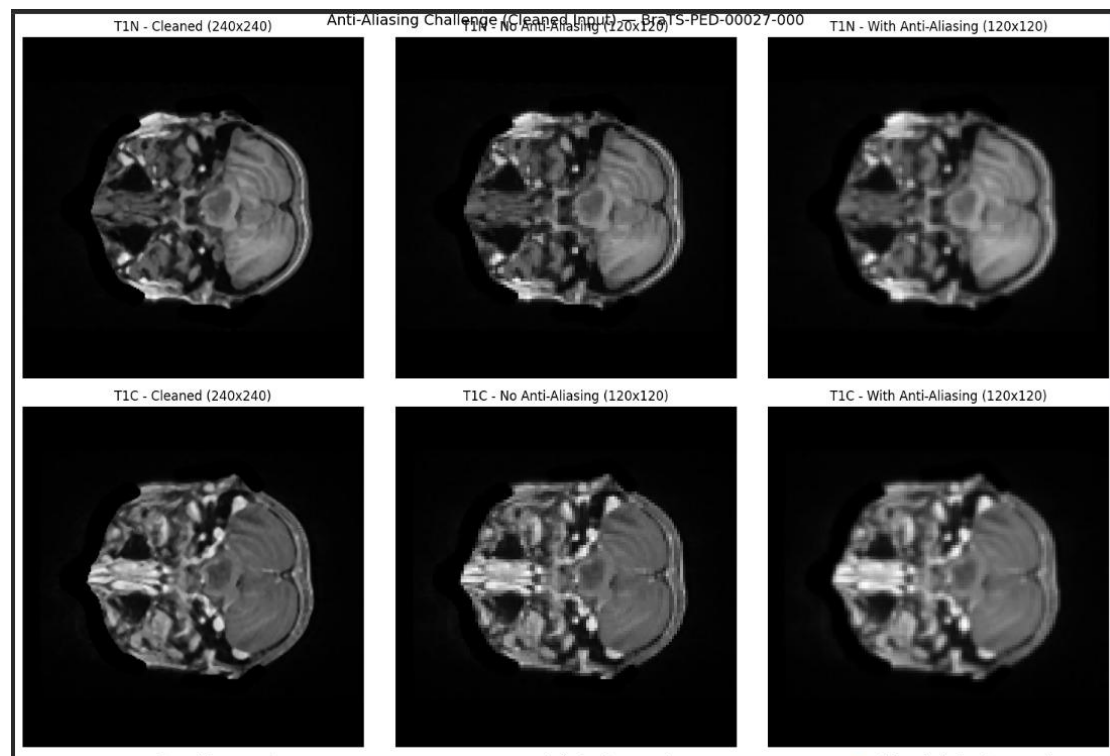
BraTS-PED-00026-000	t1c	gaussian	30.0387	0.9382
BraTS-PED-00026-000	t1c	mean	30.5955	0.9469
BraTS-PED-00026-000	t1c	median	32.8160	0.9662

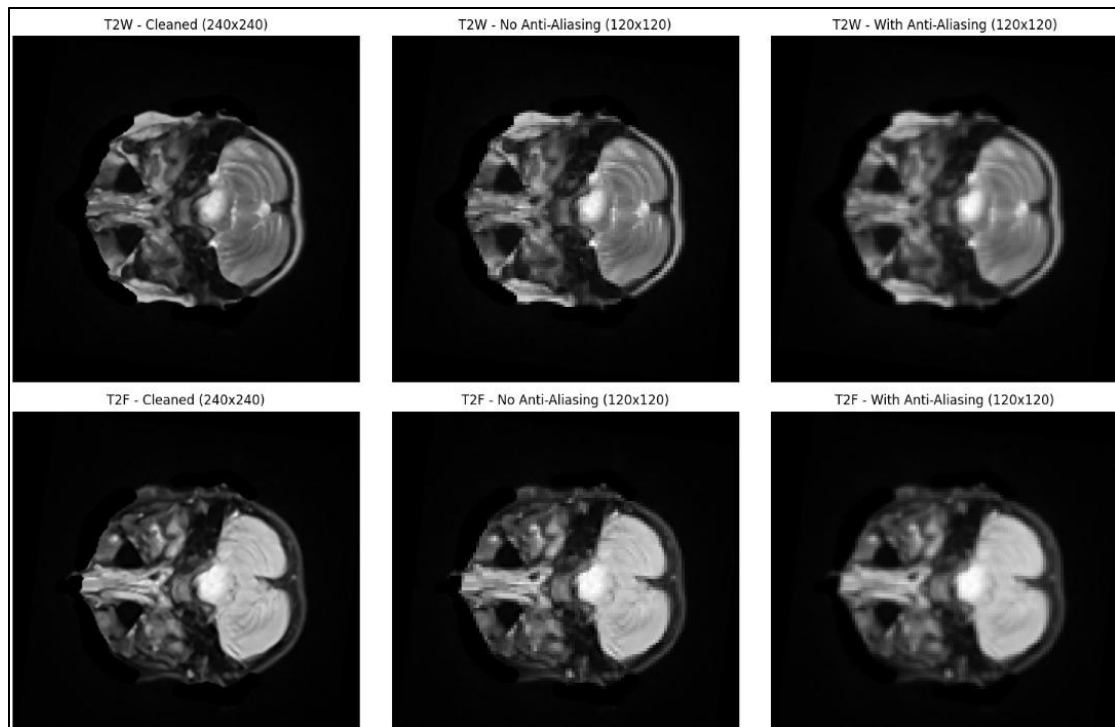
BraTS-PED-00026-000	t2w	gaussian	34.4767	0.9321
BraTS-PED-00026-000	t2w	mean	34.7431	0.9296
BraTS-PED-00026-000	t2w	median	35.7997	0.9372

BraTS-PED-00026-000	t2f	gaussian	33.8338	0.9433
BraTS-PED-00026-000	t2f	mean	34.1389	0.9453
BraTS-PED-00026-000	t2f	median	35.6651	0.9569

4.1.5 Anti-Aliasing Downsampling

Images are downsampled from 240x240 to 120x120 pixels to keep the computations manageable. Downsampling without a Gaussian pre-filter introduces checkerboard effect, fake sharpness due to the discarding of pixels randomly. To prevent this we apply a Gaussian pre-filter first and then downsample the image.

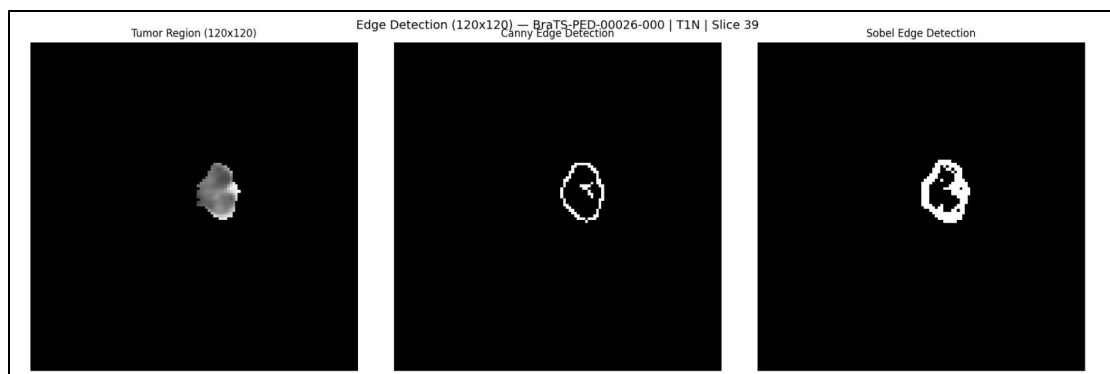




4.2 Assignment 02 - Segmentation and Boundary Analysis

4.2.1 Edge Detection (Canny and Sobel):

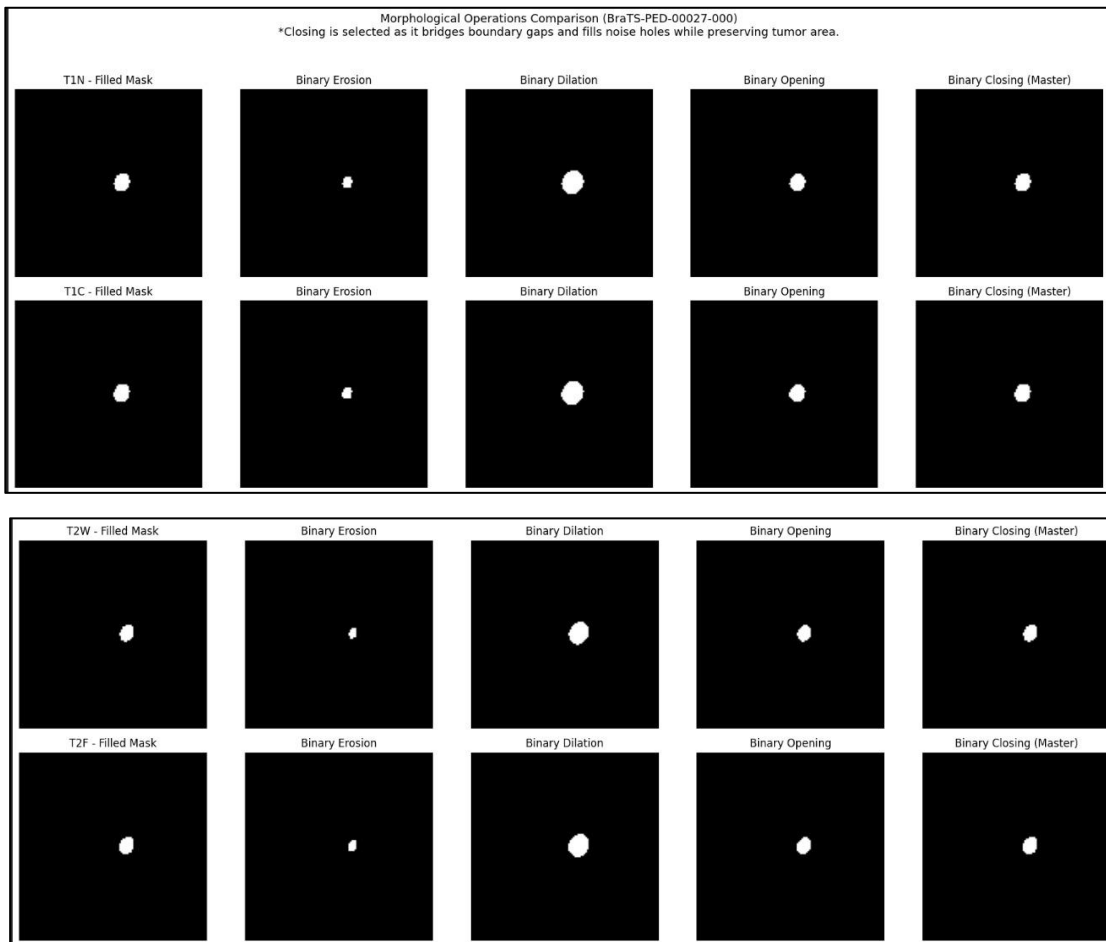
The ground truth segmentation mask from the dataset is loaded and resized to 120x120 in order to preserve binary values. Patients where the tumor region is smaller than 50 pixels are dropped. For the remaining patients, the tumor mask of each MRI slice is extracted and then edge detection is applied to that region. Canny produced edges that are kept as primary masks because they were more clean with precise boundary lines as compared to Sobel.



4.2.2 Morphological Processing:

The canny filter produces a thin broken boundary. So firstly edge thickening is applied to it then all the morphological operations using a structuring element of radius 2. Binary closing was selected as the main mask because it bridges

the gap in boundary and fills small holes without expanding the tumor boundary.

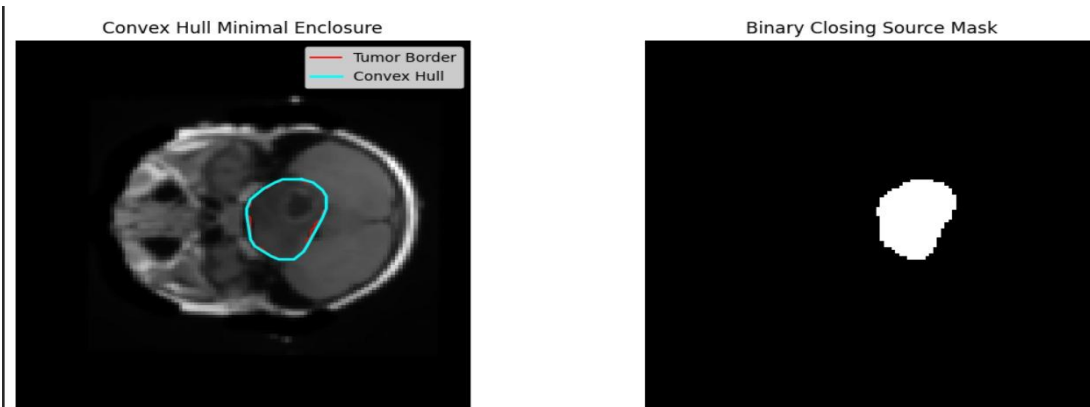


4.2.3 Chain coding and Shape Descriptor:

The cleaned binary mask is used to extract tumor boundary as pixel coordinates. Each tumor boundary is made using 8-chain code method and then its first difference is calculated to make it rotation invariant. Additionally, shape number is computed by finding the smallest circulation of first difference.

4.2.4 Convex Hull:

The smallest convex polygon that encloses the tumor region was computed for all the tumor regions in all the images. An irregular convex hull means the tumor is more spread out and irregular and it will have a large gap in its actual area and the hull. While a smooth convex hull means the tumor is regular and there is less randomness.



4.3 Assignment 03 - Feature Extraction and Classification

4.3.1 GLCM, Texture, and Geometric Features:

Two features are extracted for each tumor. For texture, we computed GLCM after quantizing the pixel values to 16 gray levels because MRI data often has high bit-depth (12-bit or 16-bit), we must first quantize the tumor pixels to a smaller number of gray levels (usually 8 or 16). If we use 256 levels, the matrix becomes too sparse and loses its statistical power. Statistical features like energy, contrast, and entropy were calculated. Geometrical features like area, circularity were calculated.

	Patient	Modality	Slice_Idx	Area	Centroid_Y	Centroid_X	Perimeter	Circularity	Energy	Contrast	Entropy
0	BraTS-PED-00026-000	T1N	39	258.0	54.251938	68.802326	59.698485	0.909710	0.980408	0.136275	0.228680
1	BraTS-PED-00026-000	T1C	39	224.0	53.718750	68.361607	56.041631	0.896265	0.982918	0.119608	0.217973
2	BraTS-PED-00026-000	T2W	39	215.0	54.358140	68.176744	53.455844	0.945492	0.983618	0.125070	0.212713
3	BraTS-PED-00026-000	T2F	39	241.0	54.273859	68.580913	57.698485	0.909700	0.981662	0.218347	0.218701
4	BraTS-PED-00027-000	T1N	27	92.0	59.086957	68.119565	34.142136	0.991782	0.992792	0.044118	0.084846

4.3.2 Classification with K-means and KNN:

There was no ground truth available for the dataset, so we use K-means ($k=2$) to assign labels first. All the computed features are normalized before clustering. The clusters with higher entropy and less circularity indicate the tumor is spread out, large and is random, labeled as malignant. The clusters with less entropy and more circularity means the tumor region is small and regular shaped, labeled as benign. These labels are then used to train a K-nearest classifier ($k=3$) and evaluated using accuracy, precision, recall, and F1 score.

	Patient	Modality	Entropy	Circularity	Label
0	BraTS-PED-00026-000	T1N	3.114139	0.882361	Malignant
1	BraTS-PED-00026-000	T1C	2.946461	0.889733	Malignant
2	BraTS-PED-00026-000	T2W	2.019479	0.814997	Malignant
3	BraTS-PED-00026-000	T2F	2.678606	0.106728	Benign
4	BraTS-PED-00027-000	T1N	3.141721	0.885686	Malignant
5	BraTS-PED-00027-000	T1C	1.491310	0.019839	Benign
6	BraTS-PED-00027-000	T2W	2.763916	0.889449	Malignant
7	BraTS-PED-00027-000	T2F	2.908141	0.224315	Benign
8	BraTS-PED-00028-000	T1N	2.107762	0.878483	Malignant
9	BraTS-PED-00028-000	T1C	2.405525	0.873736	Malignant

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=== 5-Fold KNN Cross-Validation Performance ===
Mean Accuracy: 0.9900 (± 0.0200)
Mean Precision: 0.9882 (± 0.0235)
Mean Recall: 1.0000 (± 0.0000)
Mean F1-Score: 0.9939 (± 0.0121)
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4.4 Assignment 04 - YOLOv8 Tumor Detection

4.4.1 Dataset Preparation:

The 120x120 processed images and closed morphological masks are used to build a dataset for YOLOv8. each binary mas is converted to a bounding box in YOLO format, the center x, center y width, height values are normalized. The train and test split is 80% and 20% respectively.

4.4.2 Model Training:

YOLOv8s is loaded from pre-trained weights and fine tuned on the dataset for 50 epochs. Image size is set to 128, batch size is 8, and learning rate changes from 0.01 using cosine scheduling. Bounding boxes loss is weighted at 7.5 and classification loss is 0.5.

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50 epochs completed in 0.217 hours.
Optimizer stripped from /content/drive/MyDrive/yolo_results/brain_tumor_yolo-3/weights/last.pt, 22.5MB
Optimizer stripped from /content/drive/MyDrive/yolo_results/brain_tumor_yolo-3/weights/best.pt, 22.5MB

Validating /content/drive/MyDrive/yolo_results/brain_tumor_yolo-3/weights/best.pt...
Ultralytics 8.4.60 Python-3.12.13 torch-2.11.0+cpu CPU (Intel Xeon CPU @ 2.20GHz)
Model summary (fused): 73 layers, 11,125,971 parameters, 0 gradients, 28.4 GFLOPs

```

Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%	2/2	1.9it/s	1.0s
all	20	20	0.948	0.95	0.965	0.719			

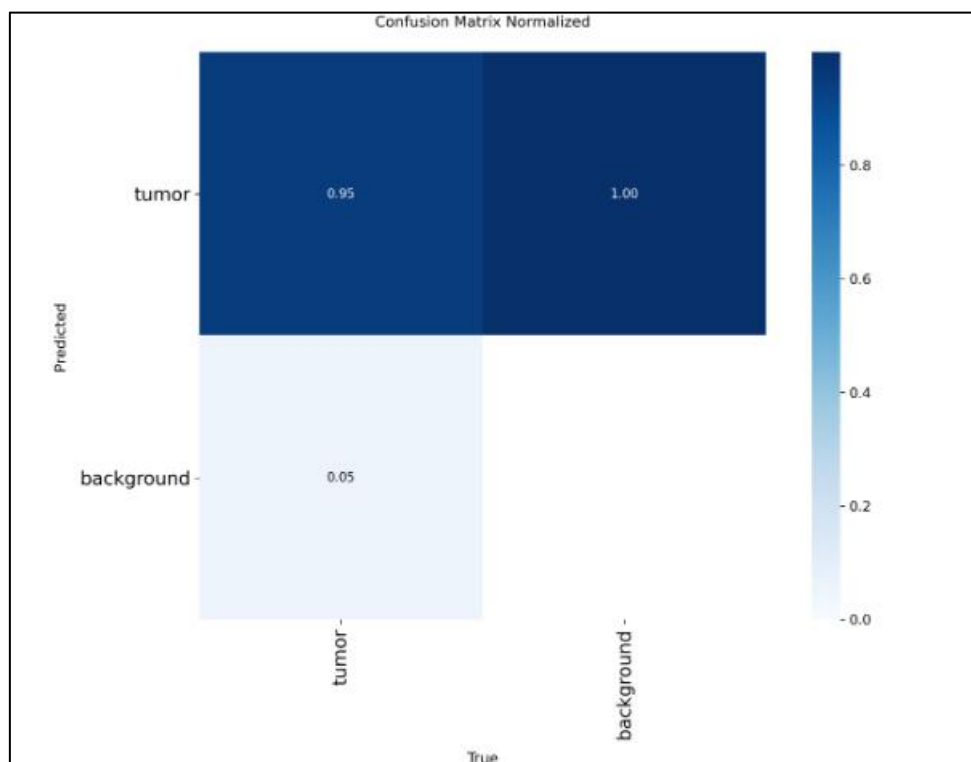
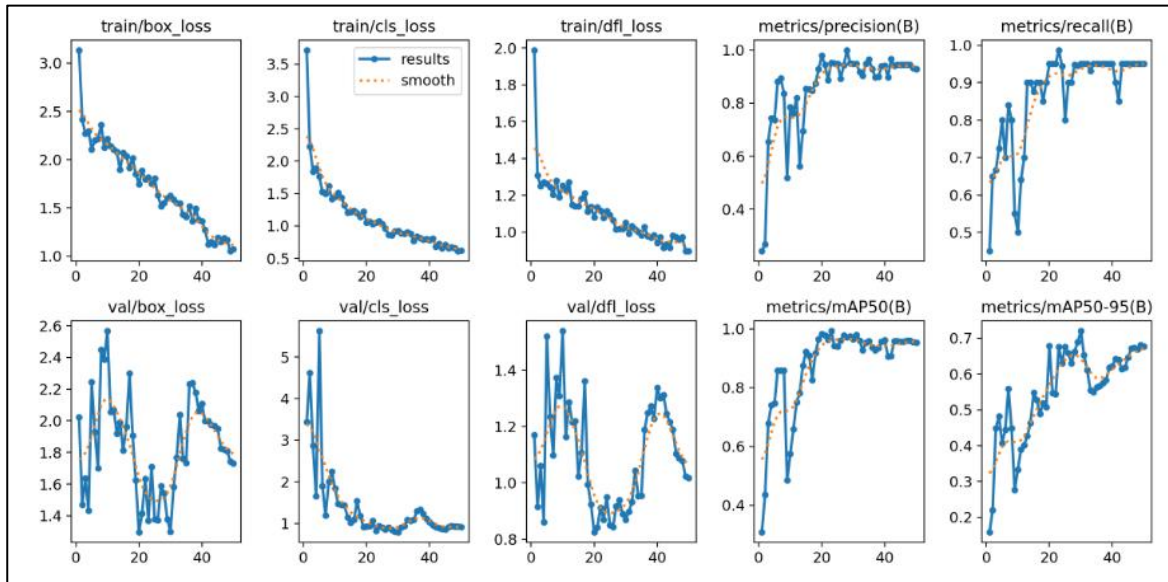
```

Speed: 0.1ms preprocess, 37.7ms inference, 0.0ms loss, 0.2ms postprocess per image
Results saved to /content/drive/MyDrive/yolo_results/brain_tumor_yolo-3
Training done!

```

4.4.3 Evaluation:

The model performance is evaluated through three outputs: training and validation loss curves over epochs, a confusion matrix, and visual prediction on a sample validation image showing the bounding box around the tumor region with confidence score.



Explanation of Confusion Matrix:

1) Top-Left (0.95) - True Positives (TP):

Out of all the validation slices that actually contain a tumor, YOLO model successfully detected and drew a bounding box around 95% of them.

2) Bottom-Left (0.05) - False Negatives (FN):

In 5% of the validation slices where a tumor was present, the model failed to detect it (meaning it predicted it as background).

3) Top-Right (1.00) - False Positives (FP):

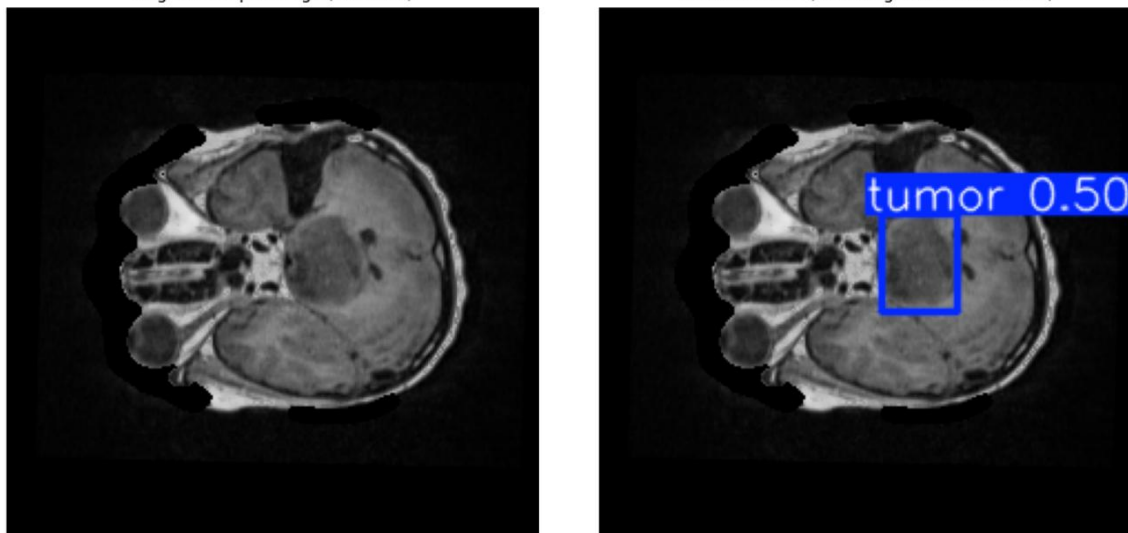
When the model looks at the empty background, it has two choices: it can make a mistake (draw a false box) or do the right thing (draw nothing). The confusion matrix only counts the mistakes the model makes on the background. It completely ignores all the times the model correctly did nothing. Because the chart only keeps track of the mistakes and ignores the correct decisions, 100% of the background data recorded on the chart are mistakes. That is why it says 1.00 (100%). It is simply showing that 100% of the background events it chose to display are mistakes.

It does not mean your model is bad or made many errors. Even if the model made only one single false box in the background, that one mistake represents 100% of the background data shown on the chart.

4) Bottom-Right (Blank) - True Negatives (TN):

This represents regions of background where the model correctly did not predict a tumor. In object detection, we do not calculate True Negatives because there is an infinite amount of background space where a model could have drawn a box but correctly chose not to. Because this value is mathematically undefined (NaN), it is left completely blank.

Example Classification Model Test: BraTS-PED-00046-000_t1c.png
Original Sample Image (120x120) YOLO Detection (Bounding Box + Confidence)



5. RESULTS & DISCUSSIONS

The automated slide selection picked the most informative brain section among all patients. Bit depth shows all scans are 16-bit images. Among the

three filter, median performed the best, its PSNR and SSIM scores were better than mean and Gaussian. Anti-aliasing reduced downsampling artifacts. Edge detection filters outlined the tumor regions perfectly. A small number of scans were dropped due to insufficient tumor area. Morphological closing consistently produced smooth and complete outlines. Chain code, first difference and shape number were computed for all combinations. Convex hull visualizations show the difference between irregular and regular tumors. GLCM features captured differences in texture across all modalities. T1c and T2FLAIR showed higher contrast because these modalities are most sensitive to tumor. K means clustering, separated the data into clusters based on entropy and circularity. The KNN classifier achieved good validation performance, showed the extracted features were helpful in separating the ground truth labels.

The trained YOLOv8s model achieved a true positive rate of 95% correctly detecting tumor region 95 times out of 100. the false negative rate was 5%. Overall, the results are strong and successful.

6. CONCLUSION

This project built a complete computer vision pipeline for brain tumor detection in pediatric MRI, starting from raw 3D volumes of MRI scans to YOLO tumor detection with confidence scores. The pipeline covered all major computer vision techniques, like intelligent slice selection, noise filtering, edge based segmentation masks, boundary encoding, feature extraction, unsupervised labeling, supervised classification and deep learning detection. Future work could expand the dataset, explore more 3D volume datasets targeting a broader region of the brain and spinal cord, replace K-means labels with expert labeled annotations.

References:

- [1] Ali, A., Li, X., Mashwani, W.K. *et al.* Computer vision based efficient segmentation and classification of multi brain tumor using computed tomography images. *Sci Rep* **15**, 32198 (2025). <https://doi.org/10.1038/s41598-025-16825-5>
- [2] A. M. Alm and S. S. Abu-Naser, "Detection of Brain Tumor Using Deep Learning," *International Journal of Academic Engineering Research (IJAER)*, vol. 6, no. 3, pp. 29–47, 2022.